



CMPT 413/713: Natural Language Processing

Text Classification - Naive Bayes

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Adapted from slides from Danqi Chen, Karthik Narasimhan, Dan Jurafsky, and Anoop Sarkar

Text classification: the task

- Inputs:

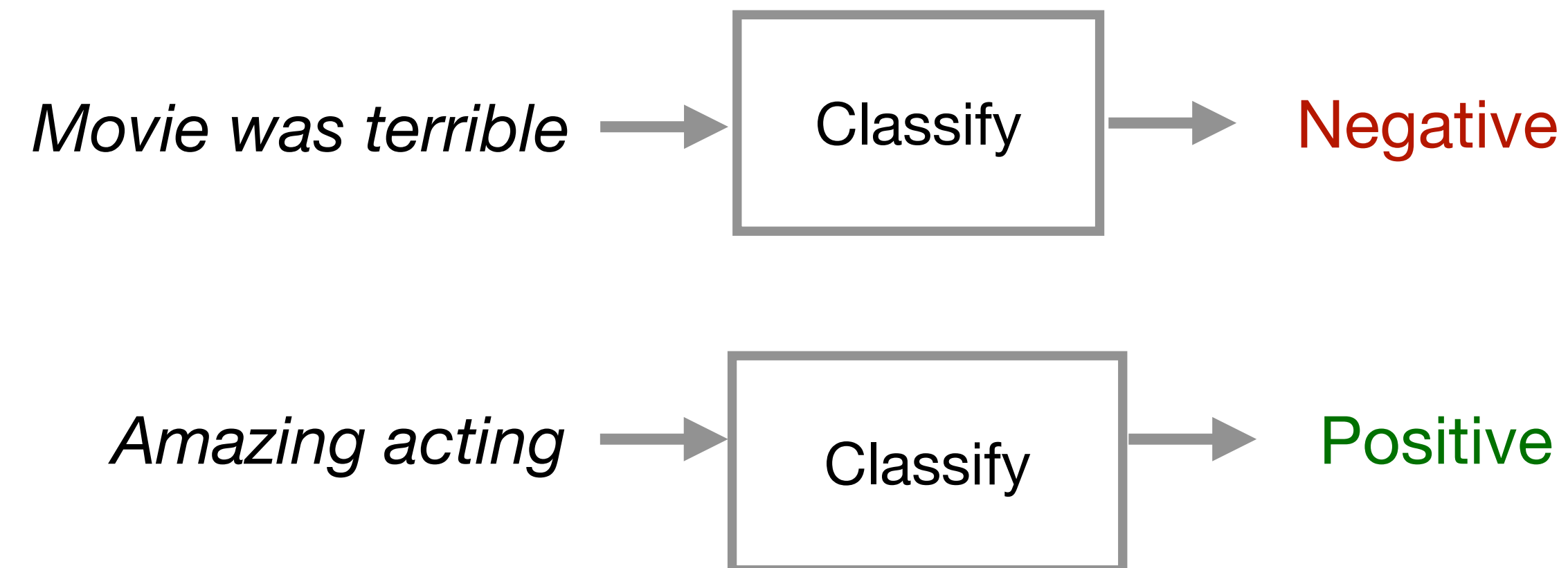
- A document **d**

- A set of classes **C** = {**c**₁, **c**₂, **c**₃, ..., **c**_m}

Multiple classes: m
Binary: m=2

- Output:

- Predicted class **c** for document **d**



Supervised Learning: Let's use statistics!

- Data-driven approach

Let the machine figure out the best patterns to use!

- Inputs:

- Set of m classes $C = \{c_1, c_2, \dots, c_m\}$
- Set of n 'labeled' documents: $\{(d_1, c_1), (d_2, c_2), \dots, (d_n, c_n)\}$

- Output: Trained classifier, $F : d \rightarrow c$

- What form should F take?
- How to learn F ?



General guidelines for model building

Two steps to building a probability model:

1. Define the model
 - What form should F take?
 - What independence assumptions do we make?
 - What are the model parameters (probability values)?
2. Estimate the model parameters (training/learning)
 - How to learn F ? What to optimize? What is the training objective?

Naive Bayes Classifier

General setting

- Let the **input** x be represented as r features: $f_j, 1 \leq j \leq r$
- Let y be the output classification
- We can have a simple classification model using **Bayes rule**

$$\begin{array}{c} \text{Posterior} \\ P(y | x) = \frac{\overset{\text{Prior}}{P(y)} \cdot \overset{\text{Likelihood}}{P(x | y)}}{\underset{\text{Evidence}}{P(x)}} \end{array}$$

- Make strong (naive) **conditional independence assumptions**

$$P(x | y) = \prod_{j=1}^r P(f_j | y) \xrightarrow{\text{Bayes rule}} P(y | x) \propto P(y) \cdot \prod_{j=1}^r P(f_j | y)$$

Naive Bayes classifier for text classification

- For text classification: **input** x is document $d = (w_1, \dots, w_k)$
- Use as our features the words w_j , $1 \leq j \leq |V|$ where V is our vocabulary
- c is the output classification
- Predicting the best class:

maximum a posteriori
(MAP) estimate

$$\begin{aligned} C_{\text{MAP}} &= \arg \max_{c \in C} P(c | d) \\ &= \arg \max_{c \in C} \frac{P(c)P(d | c)}{P(d)} \\ &= \arg \max_{c \in C} P(c)P(d | c) \end{aligned}$$

$P(d | c) \rightarrow$ Conditional probability of
generating document d from class c

$P(c) \rightarrow$ Prior probability of class c

How to represent $P(d | c)$?

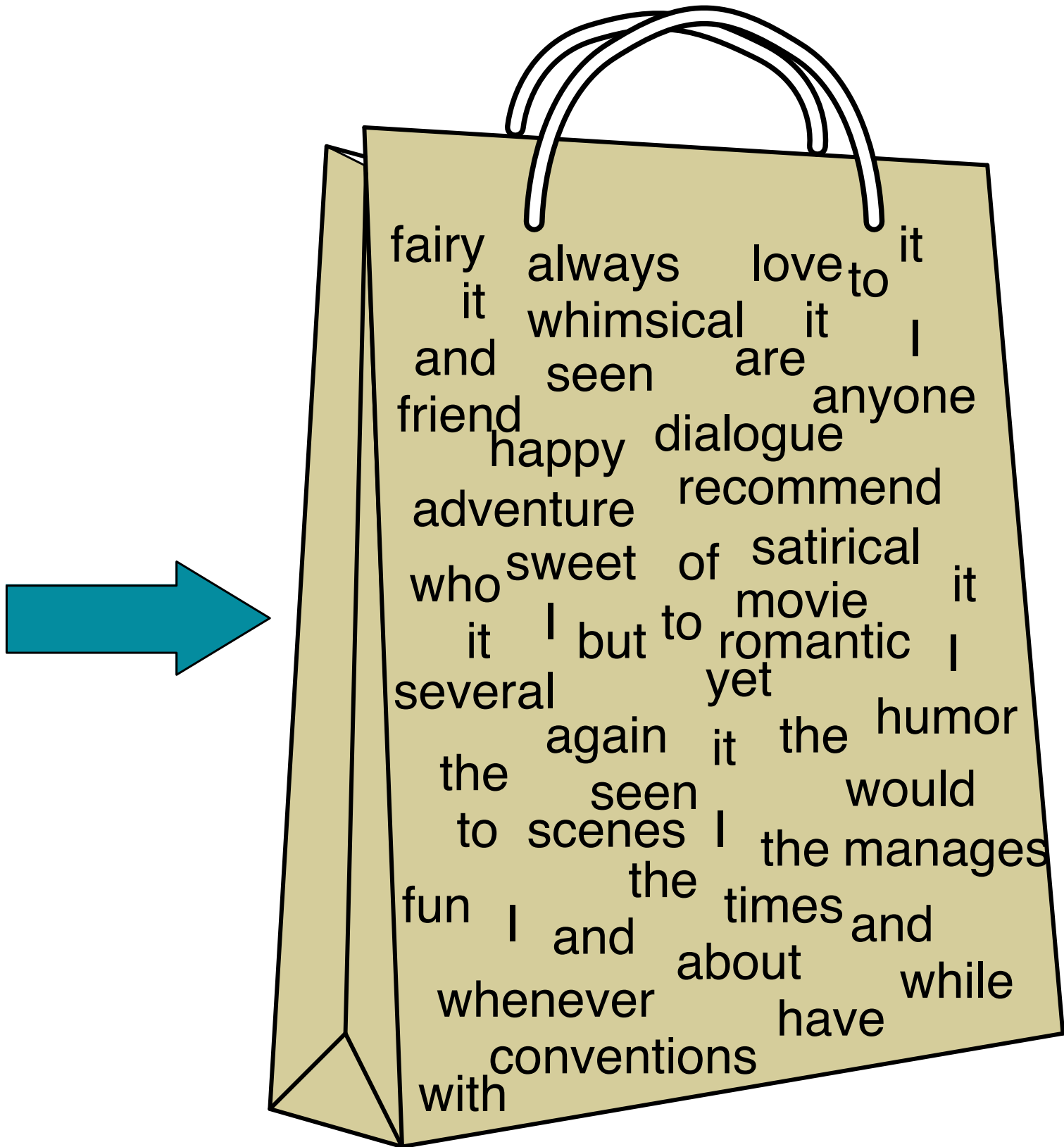
- Option 1: represent the entire sequence of words
 - $P(w_1, w_2, w_3, \dots, w_k | c)$ *(too many sequences!)*
- Option 2: **Bag of words**
 - Assume position of each word is irrelevant (both absolute and relative)
 - $P(w_1, w_2, w_3, \dots, w_k | c) = P(w_1 | c)P(w_2 | c) \dots P(w_k | c)$
 - Probability of each word is *conditionally independent* given class c

Order doesn't matter



Bag of words

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



word	count
it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

Predicting with Naive Bayes

- Once we assume that the position of each word is irrelevant and that the words are **conditionally independent** given class c , we have:

$$P(d | c) = P(w_1, w_2, w_3, \dots, w_k | c) = P(w_1 | c)P(w_2 | c) \dots P(w_k | c)$$

- The **maximum a posteriori** (MAP) estimate is now:

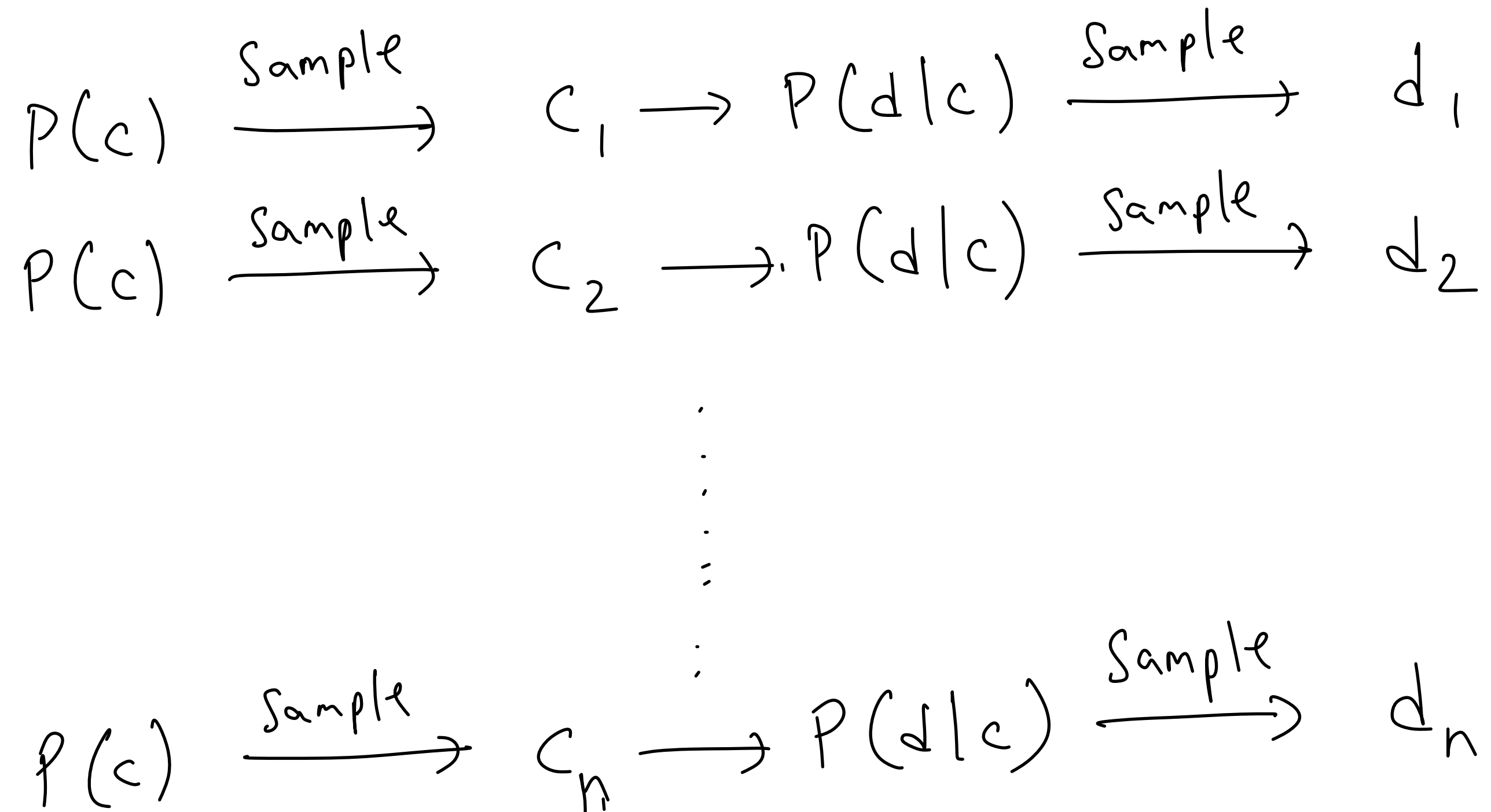
$\hat{P}$ is used to indicate the estimated probability

$$c_{\text{MAP}} = \arg \max_{c \in C} P(c)P(d | c) = \arg \max_{c \in C} \hat{P}(c) \prod_{i=1}^k \hat{P}(w_i | c)$$

Note that k is the number of tokens (words) in the document.

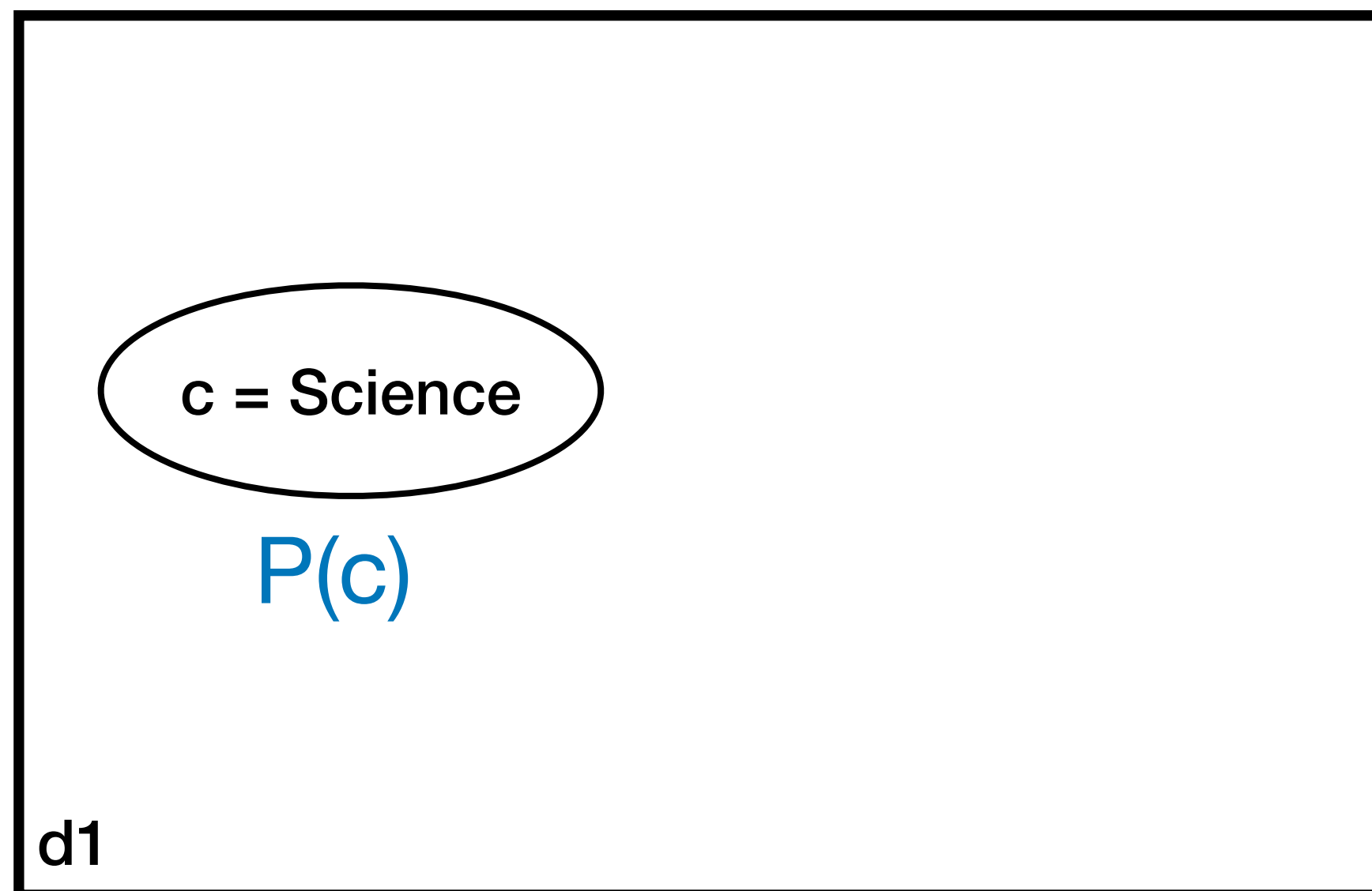
The index i is the position of the token.

Naive Bayes as a generative model



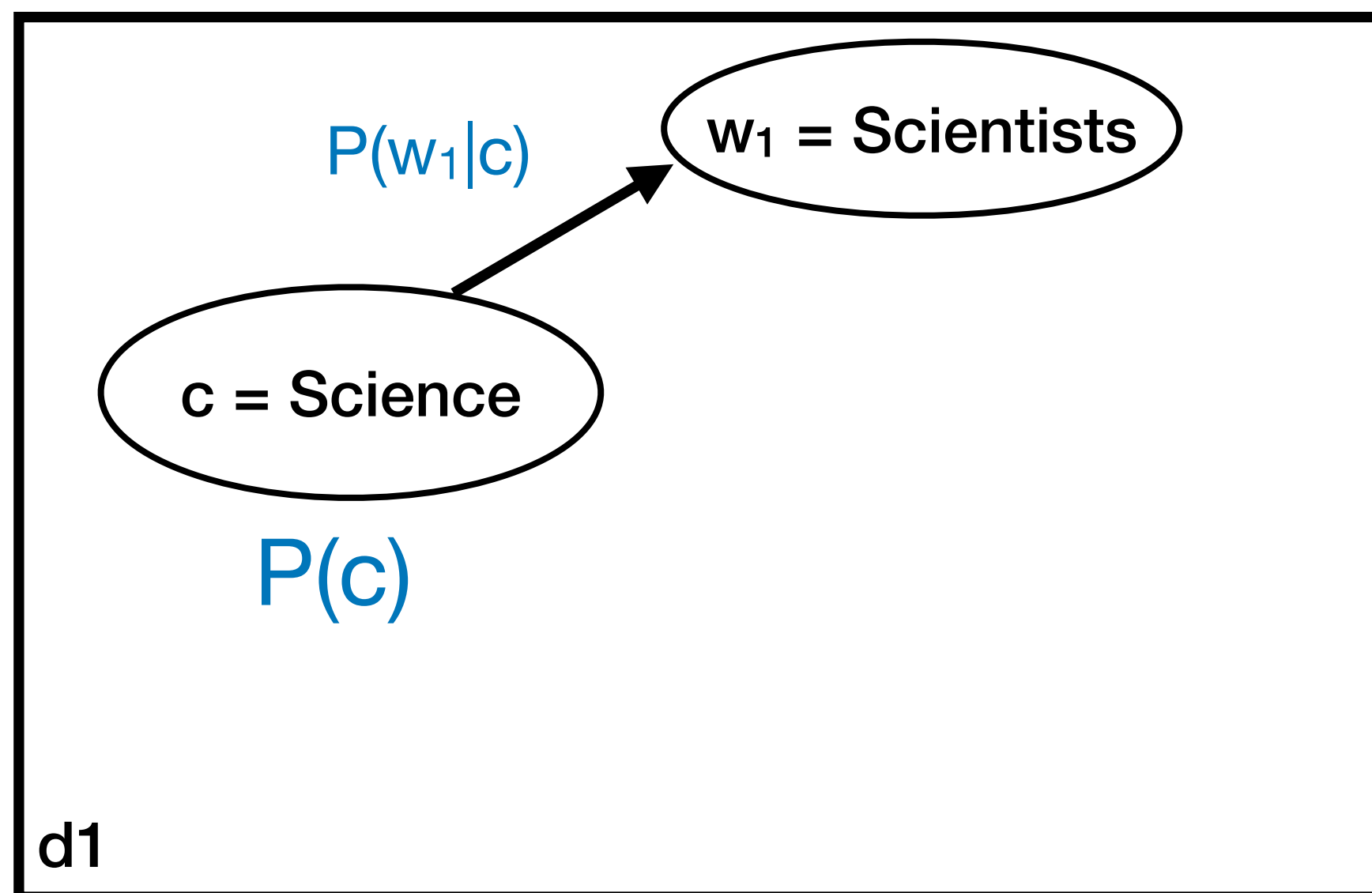
Generate the entire data set one document at a time

Naive Bayes as a generative model



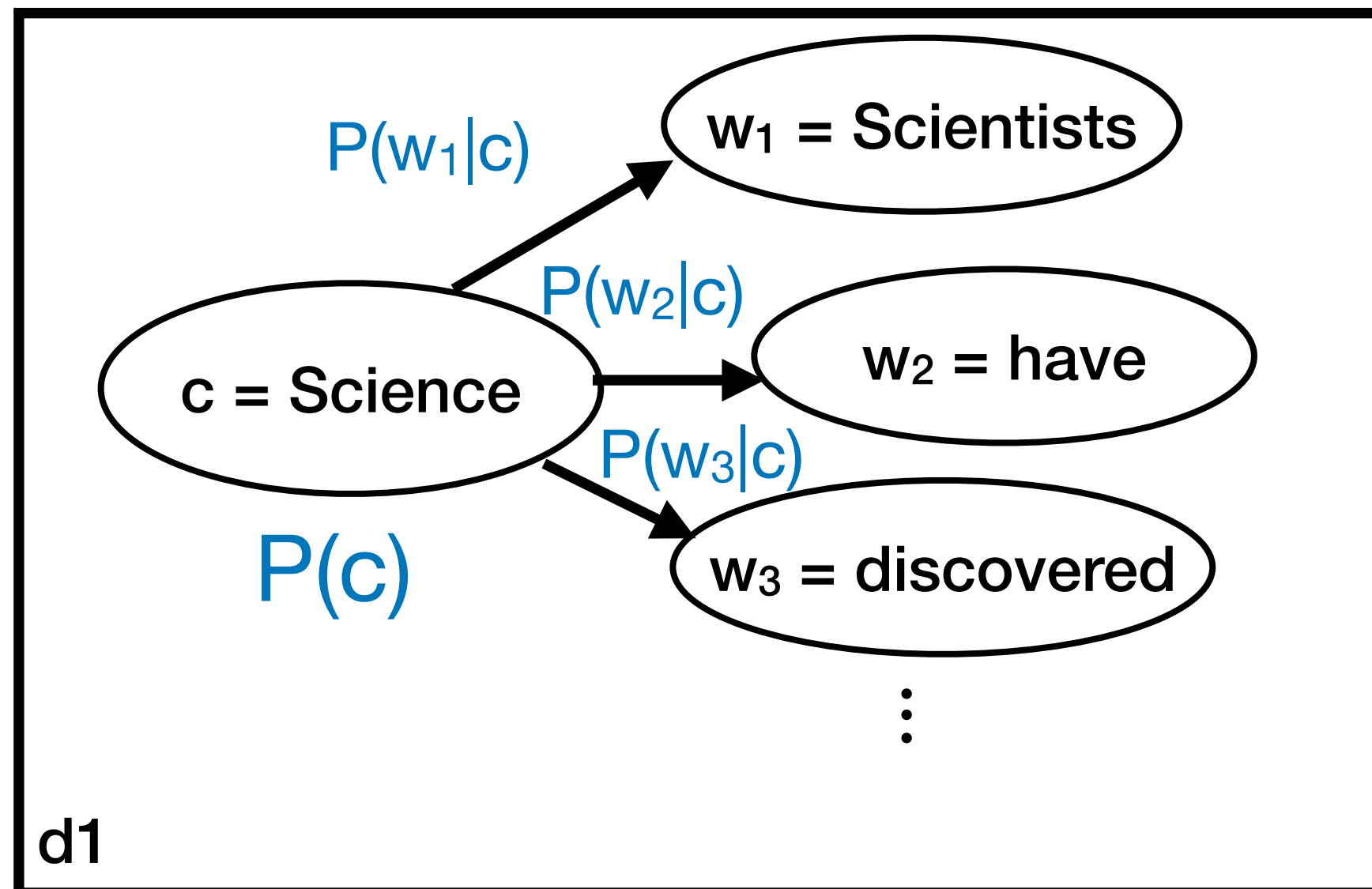
Sample a category

Naive Bayes as a generative model



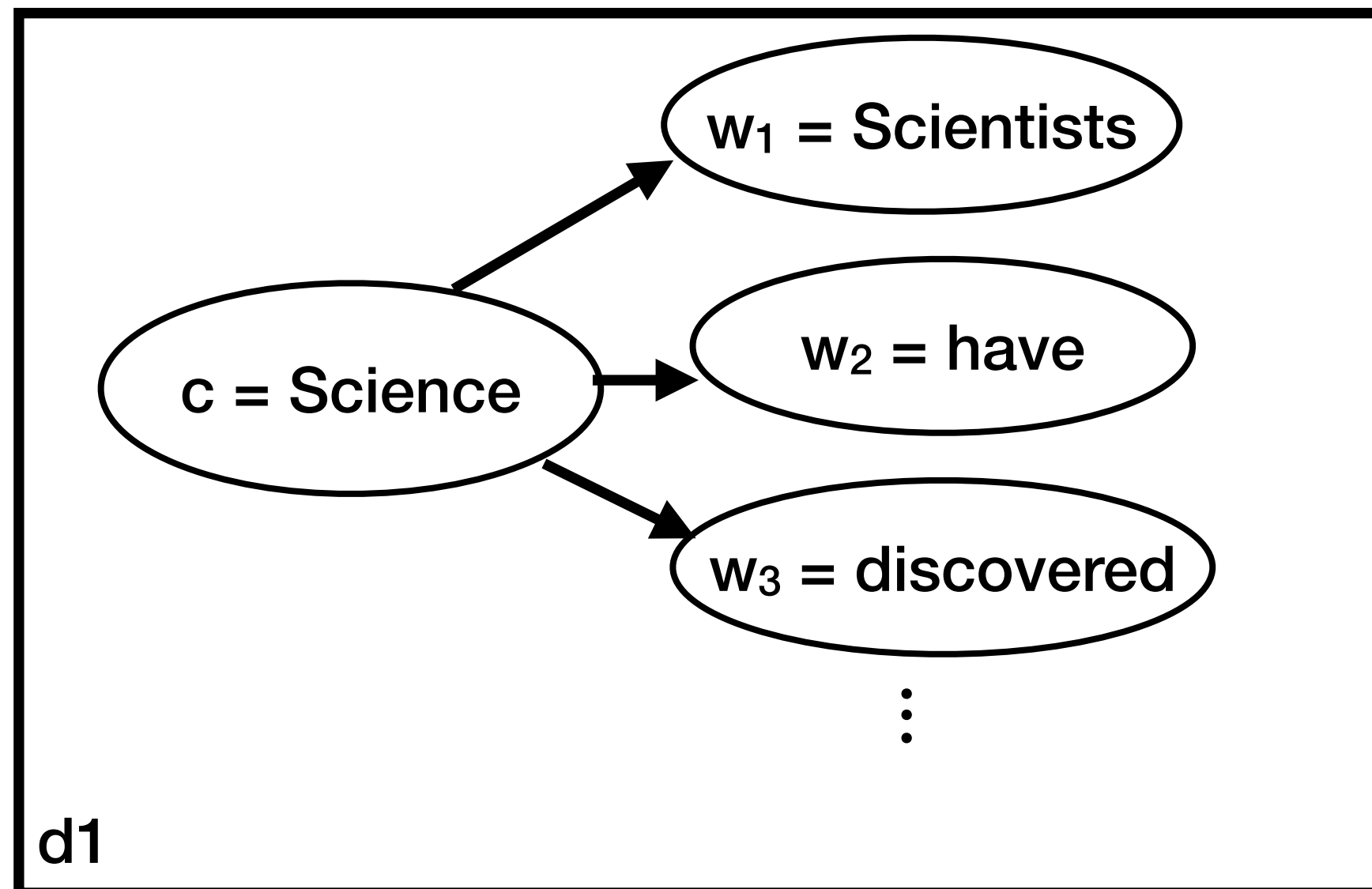
Sample words

Naive Bayes as a generative model

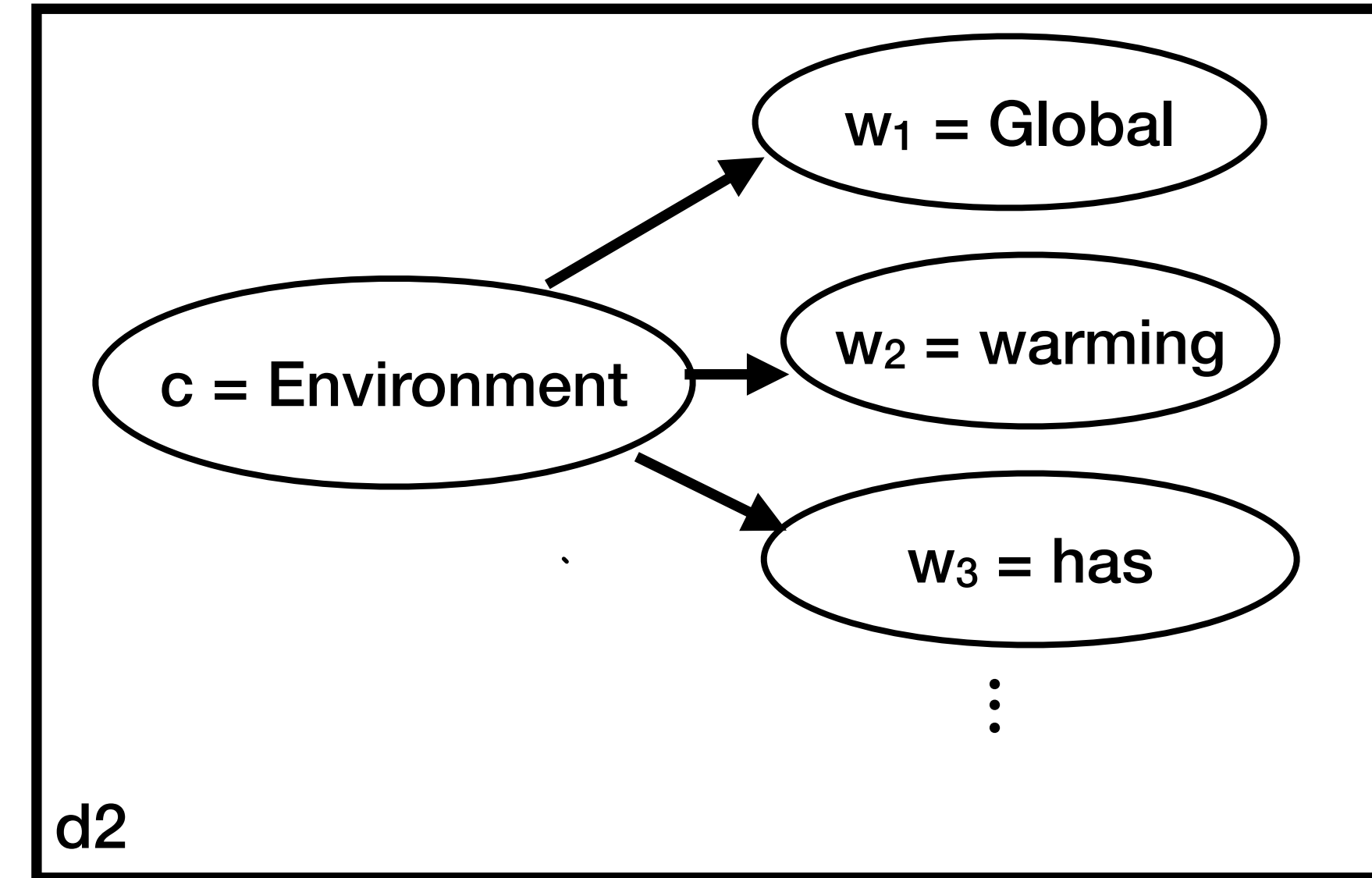


Generate the entire data set one document at a time

Naive Bayes as a generative model



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Generate the entire data set one document at a time

How to estimate probabilities

- Maximum likelihood estimate

$$\hat{P}(c_j) = \frac{\text{Count}(c_j)}{n}$$

$$\hat{P}(w_i|c_j) = \frac{\text{Count}(w_i, c_j)}{\sum_{w \in V} [\text{Count}(w, c_j)]}$$

Data sparsity

- What about when $\text{count}(\text{'amazing'}, \text{positive}) = 0$?
- Implies $P(\text{'amazing'} \mid \text{positive}) = 0$
- Given a review document, $d = \text{".... most amazing movie ever ..."}$

$$\begin{aligned} C_{\text{MAP}} &= \arg \max_{c \in C} \hat{p}(c) \prod_{i=1}^k P(w_i \mid c) \\ &= \arg \max_{c \in C} \hat{p}(c) \cdot 0 = \arg \max_{c \in C} 0 \end{aligned}$$

Can't determine the best c !

Solution: Smoothing!

- Maximum likelihood estimate

$$\hat{P}(c_j) = \frac{\text{Count}(c_j)}{n}$$

$$\hat{P}(w_i|c_j) = \frac{\text{Count}(w_i, c_j)}{\sum_{w \in V} [\text{Count}(w, c_j)]}$$

- Smoothing

$$\hat{P}(w_i|c_j) = \frac{\text{Count}(w_i, c_j) + \alpha}{\sum_{w \in V} [\text{Count}(w, c_j) + \alpha]}$$

Laplace smoothing

- Simple, easy to use
- Effective in practice

Overall process

- Input: Set of annotated documents $\{(d_i, c_i)\}_{i=1}^n$

A. Compute vocabulary $\mathbf{V}$ of all words

B. Calculate
$$\hat{P}(c_j) = \frac{\text{Count}(c_j)}{n}$$

C. Calculate
$$\hat{P}(w_i|c_j) = \frac{\text{Count}(w_i, c_j) + \alpha}{\sum_{w \in V} [\text{Count}(w, c_j) + \alpha]}$$

D. (Prediction) Given document $d = (w_1, w_2, \dots, w_k)$

$$c_{\text{MAP}} = \arg \max_c \hat{P}(c) \prod_{i=1}^k \hat{P}(w_i|c)$$

Variants

Name based on the distribution of the features

$$P(f_i|y) \rightarrow P(w_i|c)$$

Multinomial Naive Bayes

Normal counts (0,1,2,...) for each document

$$\hat{P}(c_j) = \frac{\text{Count}(c_j)}{n}$$

Binary Multinomial NB

Binarized counts (0/1) for each document

Multivariate Bernoulli NB

Estimate $P(w|c)$ as fraction of documents of class c with word w

- Explicitly model $P(!w|c) = 1 - P(w|c)$

Some work show this works better than full counts or the Multivariate Bernoulli NB

Naive Bayes Example

Smoothing with $\alpha = 1$

$$\hat{P}(c) = \frac{N_c}{N}$$
$$\hat{P}(w | c) = \frac{\text{count}(w, c) + 1}{\text{count}(c) + |V|}$$

	Doc	Words	Class
Training	1	Chinese Beijing Chinese	c
	2	Chinese Chinese Shanghai	c
	3	Chinese Macao	c
	4	Tokyo Japan Chinese	j
Test	5	Chinese Chinese Chinese Tokyo Japan	?

Some details

- **Vocabulary** is important
- **Tokenization** matters: it can affect your **vocabulary**
 - Tokenization = how you break your sentence up into tokens / words
 - Make sure you are consistent with your tokenization!

▶ *aren't* aren't
arent
are n't
aren t

▶ Emails, URLs, phone numbers, dates, emoticons

- Special multi-word tokens: NOT_happy

Some details

- **Vocabulary** is important
 - **Tokenization** matters: it can affect your **vocabulary**
 - Tokenization = how you break your sentence up into tokens / words
 - Make sure you are consistent with your tokenization!
 - Handling **unknown** words in test not in your training vocabulary?
 - Remove them from your test document! Just ignore them.
 - Handling **stop** words (common words like *a*, *the* that may not be useful)
 - Remove them from the training data!
 - In practice not that helpful, so use all words!
- Better to use
- Modified counts (tf-idf) that down weighs frequent, unimportant words
 - Better models!

Features

- In general, Naive Bayes can use **any set of features**, not just words
- URLs, email addresses, Capitalization, ...
- Domain knowledge can be crucial to performance

*Top features
for
Spam detection*

Rank	Category	Feature	Rank	Category	Feature
1	Subject	Number of capitalized words	1	Subject	Min of the compression ratio for the bz2 compressor
2	Subject	Sum of all the character lengths of words	2	Subject	Min of the compression ratio for the zlib compressor
3	Subject	Number of words containing letters and numbers	3	Subject	Min of character diversity of each word
4	Subject	Max of ratio of digit characters to all characters of each word	4	Subject	Min of the compression ratio for the lzw compressor
5	Header	Hour of day when email was sent	5	Subject	Max of the character lengths of words
(a)			(b)		
Spam URLs Features					
1	URL	The number of all URLs in an email	1	Header	Day of week when email was sent
2	URL	The number of unique URLs in an email	2	Payload	Number of characters
3	Payload	Number of words containing letters and numbers	3	Payload	Sum of all the character lengths of words
4	Payload	Min of the compression ratio for the bz2 compressor	4	Header	Minute of hour when email was sent
5	Payload	Number of words containing only letters	5	Header	Hour of day when email was sent

Naive Bayes and Language Models

- If features = bag of words, NB gives a per class unigram language model!
- For class c , assigning each word: $P(w | c)$
assigning sentence: $P(s | c) = \prod_{w \in s} P(w | c)$

Example with
positive and
negative
sentiments

Model pos	
0.1	I
0.1	love
0.01	this
0.05	fun
0.1	film

<u>I</u>	<u>love</u>	<u>this</u>	<u>fun</u>	<u>film</u>
0.1	0.1	0.01	0.05	0.1

$$P(s | pos) = 0.00000005$$

Naive Bayes as a language model

- Which class assigns the higher probability to s?

Model pos		Model neg						
0.1	I	0.2	I	<u>I</u>	<u>love</u>	<u>this</u>	<u>fun</u>	<u>film</u>
0.1	love	0.001	love	0.1	0.1	0.01	0.05	0.1
0.01	this	0.01	this	0.2	0.001	0.01	0.005	0.1
0.05	fun	0.005	fun	$P(s \text{pos}) > P(s \text{neg})$				
0.1	film	0.1	film					

Advantages of Naive Bayes

- Very Fast, low storage requirements
- Robust to Irrelevant Features
 - Irrelevant Features cancel each other without affecting results
- Very good in domains with many equally important features
 - Decision Trees suffer from *fragmentation* in such cases – especially if little data
- Optimal if the independence assumptions hold: If assumed independence is correct, then it is the Bayes Optimal Classifier for problem
- A good dependable baseline for text classification
 - **But we will see other classifiers that give better accuracy**

Failings of Naive Bayes (I)

- Independence assumptions are too strong

x1	x2	Class: $x_1 \text{ XOR } x_2$
1	1	0
0	1	1
1	0	1
0	0	0

- XOR problem: Naive Bayes cannot learn a decision boundary

Independence assumption broken!

- Both variables are jointly required to predict class

Failings of Naive Bayes (2)

- Class imbalance:

Does not handle rare classes well

- One or more classes have more instances than others

- Okay if test distribution follows training and you don't care about the rare classes

- Data skew causes NB to prefer one class over the other

- Low macro-average metrics

Re-weight classes if needed

- 100 documents with class=MA and "Boston" occurring once each

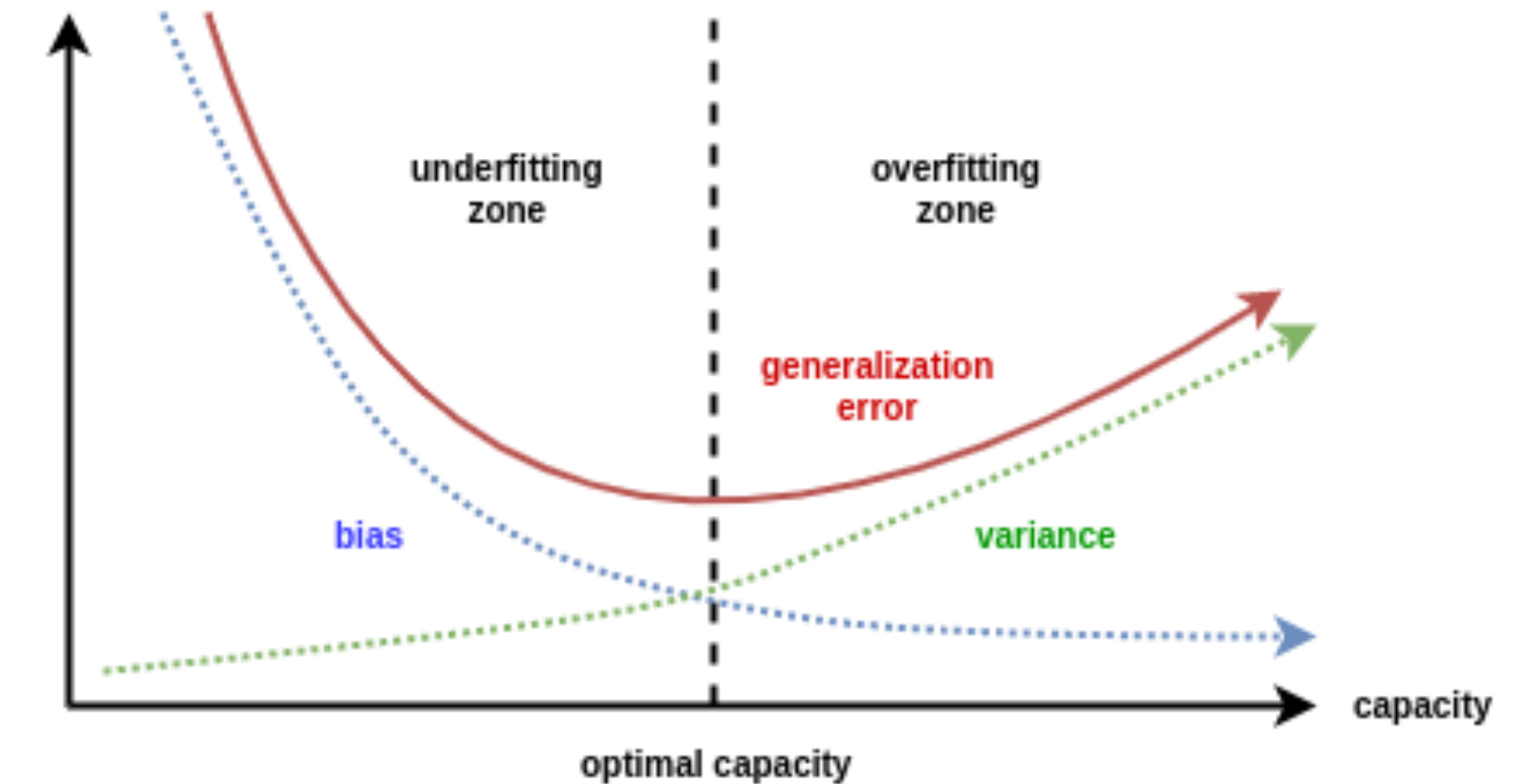
- 10 documents with class=BC and "Vancouver" occurring once each

- New document d : "Boston Boston Vancouver Vancouver Vancouver"

$$P(\text{class} = \text{MA} \mid d) > P(\text{class} = \text{BC} \mid d)$$

When to use Naive Bayes

- Small data sizes:
 - Naive Bayes is great! (high bias)
 - Rule-based classifiers might work well too
- Medium size datasets:
 - More advanced classifiers might perform better (e.g. SVM, logistic regression)
- Large datasets:
 - Naive Bayes becomes competitive again (although most classifiers work well)



Practical text classification

- Domain knowledge is crucial to selecting good features
- Handle class imbalance by re-weighting classes
- Use log scale operations instead of multiplying probabilities
- Since $\log(xy) = \log(x) + \log(y)$:

$$c_{\text{MAP}} = \arg \max_{c_j \in C} \log P(c_j) + \sum_{i=1}^k \log P(x_i | c_j)$$

better to sum logs of
probabilities than to multiply
probabilities

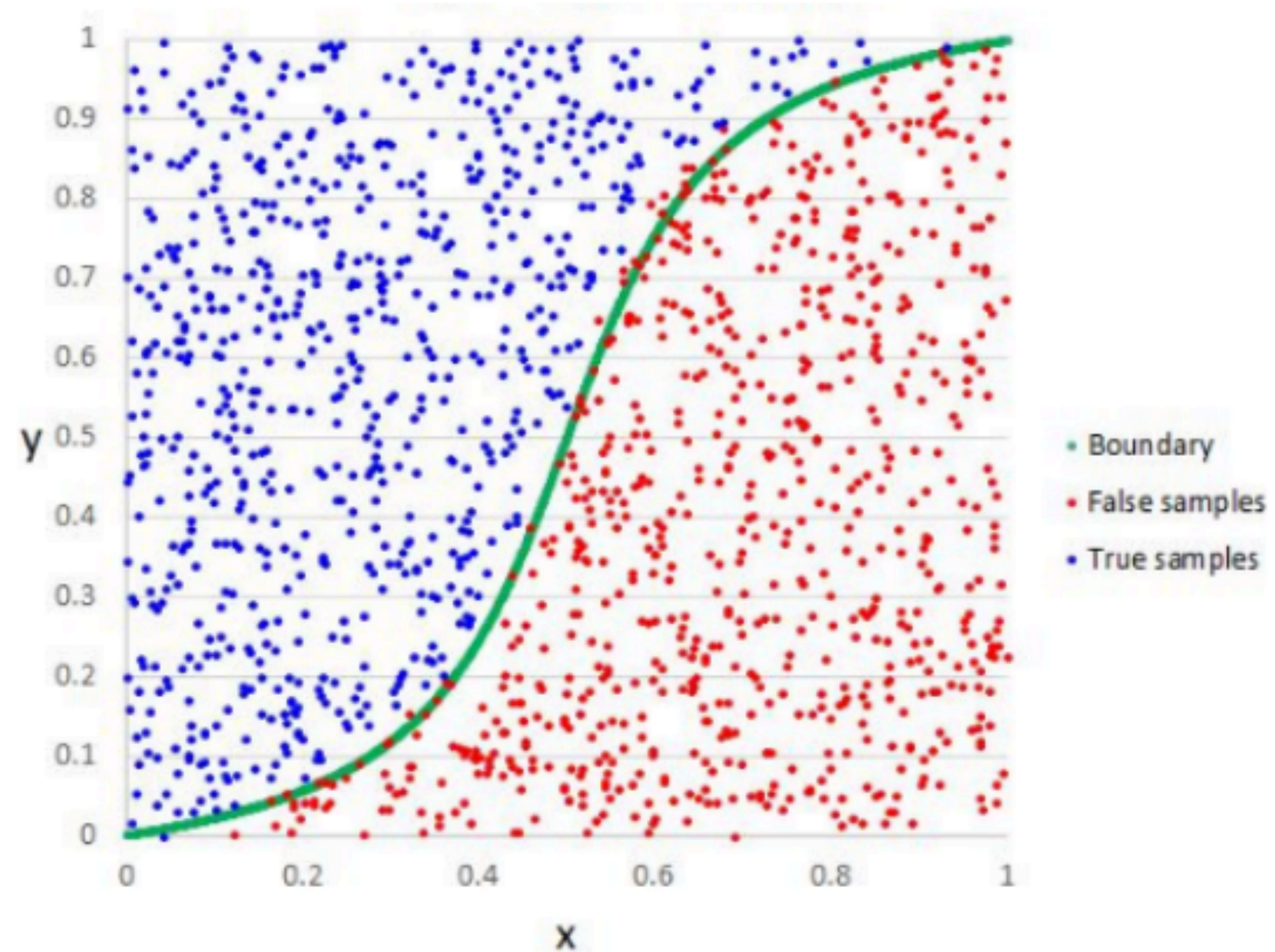
- Class with highest un-normalized log probability score is still the most probable
- Model is now just max of sum of weights

Generative vs Discriminative Models

- Naive Bayes is a Generative Model: It models $p(y | x) \propto p(y)p(x | y)$
- It models how the document is generated from words
- You can use this model to sample documents
- Next: Logistic Regression, a Discriminative model that models $p(y | x)$ directly.

Discriminative Model

- Logistic Regression: model $p(y | x)$ directly



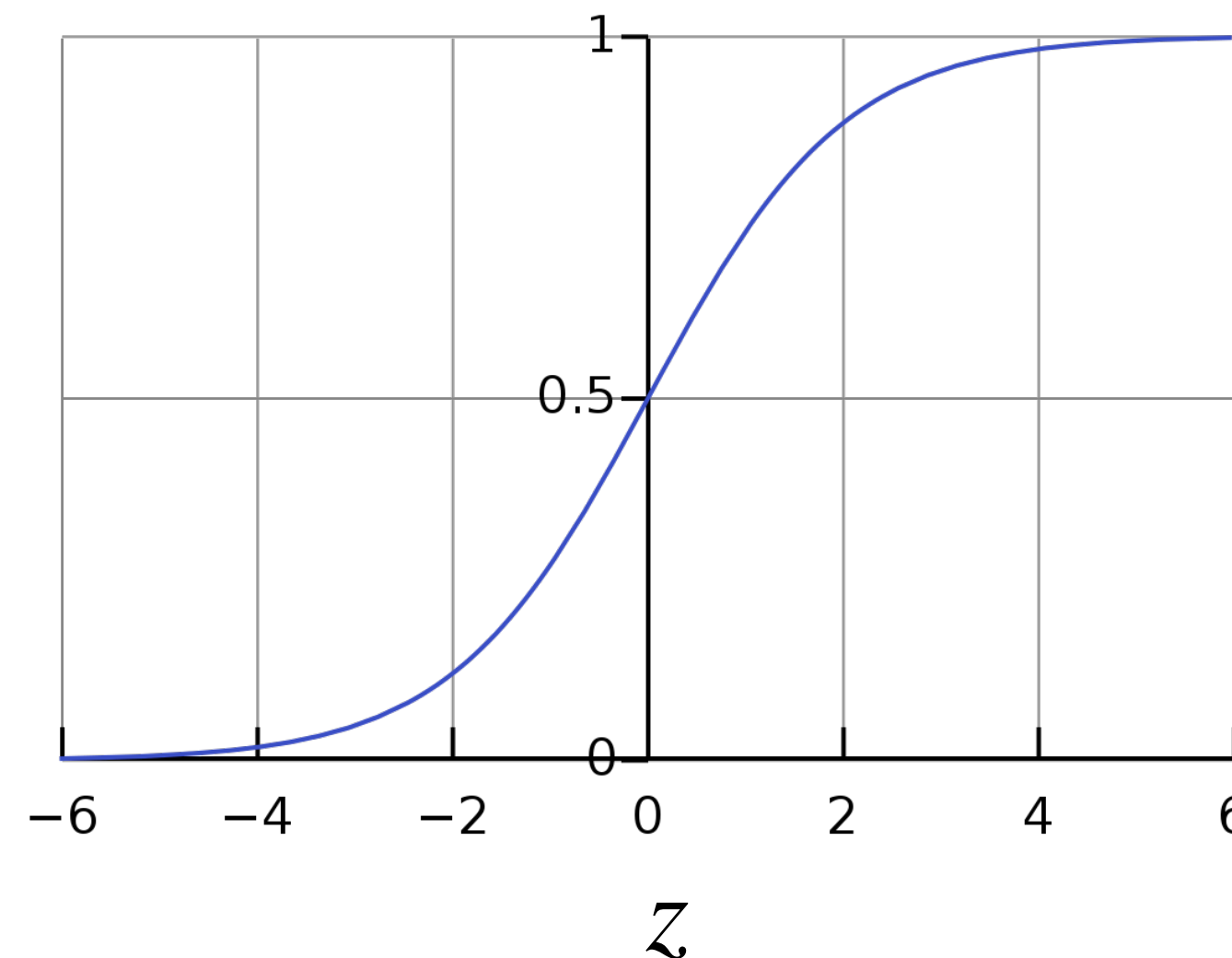
Classification function

- *Given:* Input feature vector $[x_1, x_2, \dots, x_d]$
- *Output:* $P(y = 1 | x)$ and $P(y = 0 | x)$ *(binary classification)*
- Use a *function*, $F : \mathbb{R}^d \rightarrow [0,1]$ to model the probability $P(y | x)$

- Sigmoid:

$$P(y|x) = \frac{1}{1 + e^{-z}}$$

$$z = \sum_{i=1}^d w_i x_i + b$$



$P(y|x)$